

## PROG2 Day 3: Debugging Exam

### EXAM INSTRUCTIONS

#### Rules:

- ✓ You may use pen and paper for tracing
- ✗ You may NOT use a compiler or IDE
- ✗ You may NOT ask for help from others

- ✓ Show ALL your work - line numbers and explanations

#### Answer on separate paper:

- Label each answer clearly (Q1, Q2, Q3, Q4)
- Show all work for full credit

60 min | 100 points (25 each)

### Question 1

25 points

#### CONTEXT

This program should count how many numbers between 1 and 20 are positive AND greater than 10.

**Expected:** Numbers 11-20 should be counted (10 numbers)

**Expected output:** Count: 10

```
1 #include <stdio.h>
2
3 int main() {
4     int count = 0;
5
6     for (int i = 1; i <= 20; i++) {
7         if (i > 0 || i > 10) {
8             count++;
9         }
10    }
11
12    printf("Count: %d\n", count);
13
14    return 0;
15 }
```

⚠ **Actual output:** Count: 20

#### YOUR TASK

- Find the bug:** What line(s) have the error?
- Explain why:** Why does this cause the wrong output?
- Trace it:** Show what happens with 3 test values ( $i = 5$ ,  $i = 15$ ,  $i = 1$ )
- Write the fix:** What should the code be?
- Verify:** Show the corrected code produces Count: 10

### Question 2

25 points

#### CONTEXT

This program should calculate the factorial of 5

**Formula:**  $5! = 5 \times 4 \times 3 \times 2 \times 1 = 120$

**Expected output:**  $5! = 120$

```
1 #include <stdio.h>
2
3 int main() {
4     int result = 1;
5
6     for (int i = 1; i < 5; i++) {
7         result = result * i;
8     }
9
10    printf("5! = %d\n", result);
11
12    return 0;
13 }
14
15
```

⚠ **Actual output:**  $5! = 24$

#### YOUR TASK

- Find the bug:** What's wrong with the loop condition (line 6)?
- Trace it:** Show what gets multiplied at each iteration
- Explain why:** Why does 24 appear instead of 120?
- Write the fix:** Change the loop condition only (don't change the body)
- Verify:** Confirm your fix produces 120

### Question 3

25 points

#### CONTEXT

This program determines if a year is a leap year.

**Rules:** A year is a leap year if it's divisible by 4 AND (NOT divisible by 100 OR divisible by 400)

#### Test Cases:

- 2000: "Leap year" ✓
- 1900: "Not leap" ✗ (wrong)
- 2020: "Leap year" ✓
- 2019: "Not leap" ✓

```
1 #include <stdio.h>
2
3 int main() {
4     int year = 2000;
5
6     if (year % 4 == 0 || year % 100 != 0 || year % 400 == 0) {
7         printf("Leap year\n");
8     } else {
9         printf("Not a leap year\n");
10    }
11
12    return 0;
13 }
```

⚠ **Problem:** Year 1900 incorrectly outputs "Leap year" (should be "Not a leap year")

#### YOUR TASK

- Find the bug:** What's wrong with the condition on line 6?
- Analyze:** Why does year 1900 incorrectly print "Leap year"?
- Trace:** Test 1900 with the buggy condition step by step
- Write the fix:** Rewrite the complete if condition correctly
- Verify:** Test your fixed condition with all 4 cases

### Question 4

25 points

#### CONTEXT

This program should sum all EVEN numbers from 10 to 15

**Expected calculation:**  $10 + 12 + 14 = 36$

**Expected output:** Sum of evens: 36

```
1 #include <stdio.h>
2
3 int main() {
4     int sum = 0;
5
6     for (int i = 10; i <= 15; i++) {
7         if (i % 2 = 0) {
8             sum = sum + i;
9         }
10    }
11
12    printf("Sum of evens: %d\n", sum);
13
14    return 0;
15 }
```

⚠ **Actual status:** Code does NOT compile

#### YOUR TASK

- Find ALL bugs:** There are 2 bugs in this code. Find both!
- Explain each bug:** What's wrong and why?
- Write the fixes:** Correct both bugs
- Trace:** Show what the fixed code does for each iteration
- Verify:** Confirm the output is 36

#### TIME TIPS

Q1: ~15 min | Q2: ~12 min | Q3: ~15 min | Q4: ~15 min | Review: ~3 min

Show your work systematically. 🧸 → 🧰

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